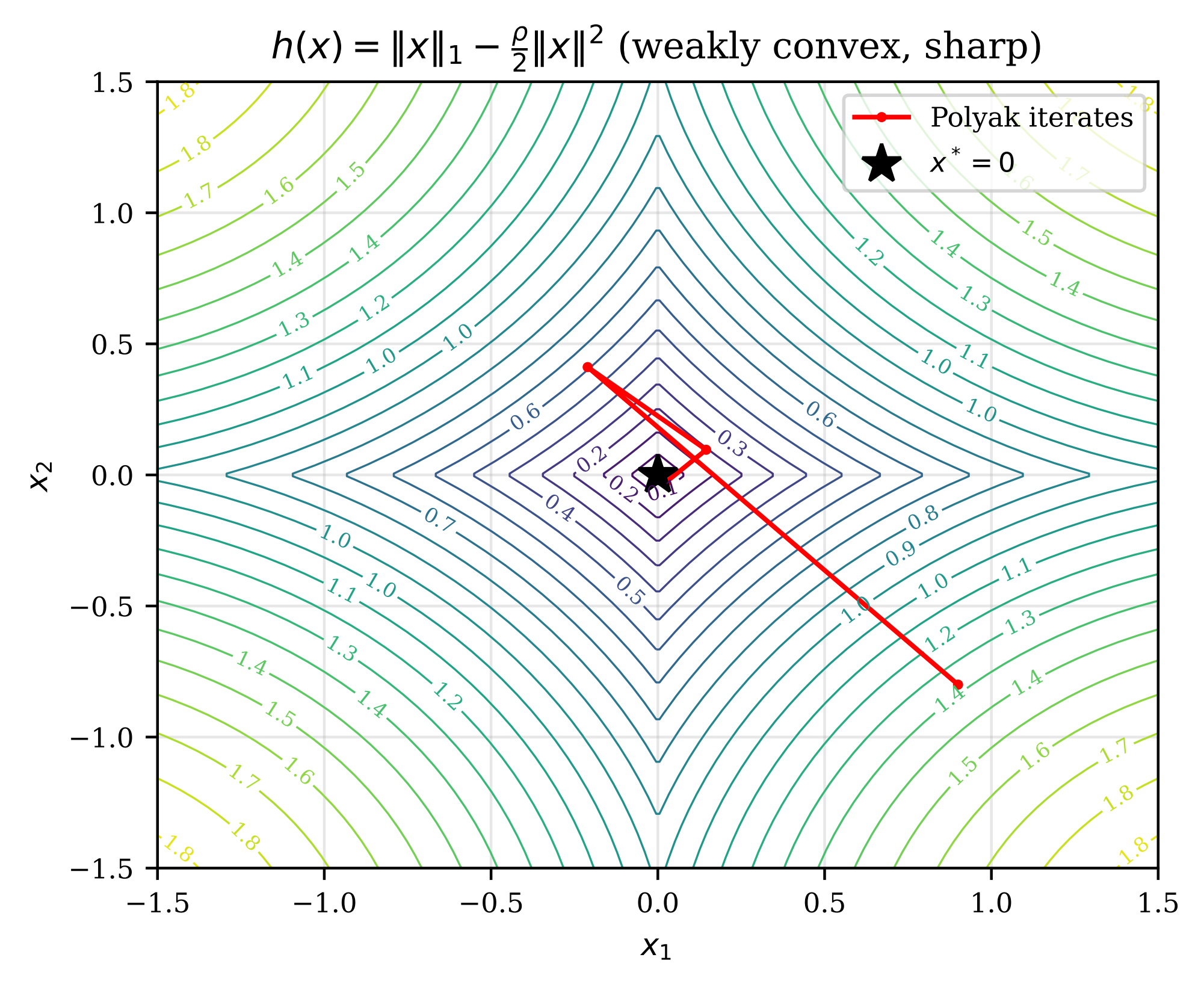
$h(x) = \|x\|_1 - \frac{\rho}{2}\|x\|^2$ (weakly convex, sharp)
Polyak iterates
$x^* = 0$
x_1
x_2
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5